

```

import os
import re
import string
import warnings
from collections import Counter

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from wordcloud import WordCloud

import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords, wordnet
from nltk.stem import WordNetLemmatizer
from nltk import pos_tag

from sklearn.feature_extraction.text import CountVectorizer
from sklearn.decomposition import LatentDirichletAllocation

warnings.filterwarnings("ignore")

# NLTK resources
for resource in [
    "punkt", "punkt_tab", "stopwords", "wordnet",
    "omw-1.4", "averaged_perceptron_tagger",
    "averaged_perceptron_tagger_eng"
]:
    try:
        nltk.download(resource, quiet=True)
    except Exception:
        pass

print("Libraries imported successfully.")

Libraries imported successfully.

CSV_PATH = "Adult_Immunization_Features_Headlines.csv"

df = pd.read_csv(CSV_PATH)

print("Dataset shape:", df.shape)
display(df.head())
print("\nColumns:")
print(df.columns.tolist())

Dataset shape: (366, 4)

```

	Date	Source
\		
0	October 2004	Medscape Infectious Diseases
1	July 2004	Journal of the American Board of Family Medicine
2	June 2004	Medscape Nurses
3	April 2004	Alimentary Pharmacology & Therapeutics
4	November 2003	Medscape Infectious Diseases

	Headlines	\
0	Infectious Diseases: October 30, 2004	
1	Influenza Vaccination in Pregnancy	
2	Bacterial Meningitis: A Deadly But Preventable...	
3	Hepatitis Vaccination in Patients With Chronic...	
4	Emerging Infections	

	Teaser
0	Dr. Bartlett reviews the latest on influenza, ...
1	Are enough family physicians and obstetricians...
2	Those whose lives have been touched by this le...
3	Despite a declining incidence of acute viral h...
4	Dr. John Bartlett reports on the latest data f...

Columns:

```
['Date', 'Source', 'Headlines', 'Teaser']
```

```
def clean_column_name(col):
    col = str(col).replace("\uffff", "").strip()
    col = re.sub(r'&#x65;', 'e', col, flags=re.I) # handles Source
    col = re.sub(r'&#x48;', 'H', col, flags=re.I) # handles Headlines
    col = re.sub(r'^A-Za-z0-9+', '_', col).strip('_').lower()
    return col
```

```
df.columns = [clean_column_name(c) for c in df.columns]
```

Standardize common variants

```
rename_map = {}
for c in df.columns:
    if c in {"headline", "headlines", "title", "titles"}:
        rename_map[c] = "headlines"
    elif c in {"teaser", "description", "summary"}:
        rename_map[c] = "teaser"
    elif c in {"source", "sources"}:
        rename_map[c] = "source"
```

```

elif c in {"date", "published_date", "publication_date"}:
    rename_map[c] = "date"

df = df.rename(columns=rename_map)

required = ["headlines", "teaser"]
missing = [c for c in required if c not in df.columns]
if missing:
    raise ValueError(f"Missing required text column(s): {missing}.
Found: {df.columns.tolist()}")

print("Normalized columns:", df.columns.tolist())
display(df.head())

```

Normalized columns: ['date', 'source', 'headlines', 'teaser']

	date	source
0	October 2004	Medscape Infectious Diseases
1	July 2004	Journal of the American Board of Family Medicine
2	June 2004	Medscape Nurses
3	April 2004	Alimentary Pharmacology & Therapeutics
4	November 2003	Medscape Infectious Diseases

	headlines
0	Infectious Diseases: October 30, 2004
1	Influenza Vaccination in Pregnancy
2	Bacterial Meningitis: A Deadly But Preventable...
3	Hepatitis Vaccination in Patients With Chronic...
4	Emerging Infections

	teaser
0	Dr. Bartlett reviews the latest on influenza, ...
1	Are enough family physicians and obstetricians...
2	Those whose lives have been touched by this le...
3	Despite a declining incidence of acute viral h...
4	Dr. John Bartlett reports on the latest data f...

```

df["headlines"] = df["headlines"].fillna("").astype(str)
df["teaser"] = df["teaser"].fillna("").astype(str)
df["text"] = (df["headlines"] + " " + df["teaser"]).str.strip()

stop_words = set(stopwords.words("english"))
custom_stopwords = {
    "said", "says", "say", "news", "new", "also", "one", "two",
    "may", "could", "would", "get", "got", "use", "using"
}

```

```

}
stop_words.update(custom_stopwords)

lemmatizer = WordNetLemmatizer()

def penn_to_wordnet(tag):
    if tag.startswith("J"):
        return wordnet.ADJ
    if tag.startswith("V"):
        return wordnet.VERB
    if tag.startswith("N"):
        return wordnet.NOUN
    if tag.startswith("R"):
        return wordnet.ADV
    return wordnet.NOUN

def preprocess_text(text):
    text = str(text).lower()
    text = re.sub(r"http\S+|www\.\S+", " ", text)
    text = re.sub(r"^[a-z\s]", " ", text)
    text = re.sub(r"\s+", " ", text).strip()

    tokens = word_tokenize(text)
    tokens = [t for t in tokens if t not in stop_words and len(t) > 2]

    # POS-aware lemmatization
    tagged = pos_tag(tokens)
    lemmas = [
        lemmatizer.lemmatize(token, penn_to_wordnet(tag))
        for token, tag in tagged
    ]
    lemmas = [t for t in lemmas if t not in stop_words and len(t) > 2]
    return lemmas

df["tokens"] = df["text"].apply(preprocess_text)
df["clean_text"] = df["tokens"].apply(lambda x: " ".join(x))

print("Preprocessing completed.")

Preprocessing completed.

sample_cols = [c for c in ["headlines", "teaser", "clean_text"] if c
in df.columns]
display(df[sample_cols].head(10))

```

	headlines \
0	Infectious Diseases: October 30, 2004
1	Influenza Vaccination in Pregnancy
2	Bacterial Meningitis: A Deadly But Preventable...
3	Hepatitis Vaccination in Patients With Chronic...
4	Emerging Infections

```

5 New Organisms, New Drugs, New Tests, New Guide...
6           Asthma and Influenza Vaccination
7       Some Answers to Smallpox Vaccination Questions
8 Flu Vaccine for Cardiovascular Prevention in t...
9                               October 2002

```

```

                                teaser \
0 Dr. Bartlett reviews the latest on influenza, ...
1 Are enough family physicians and obstetricians...
2 Those whose lives have been touched by this le...
3 Despite a declining incidence of acute viral h...
4 Dr. John Bartlett reports on the latest data f...
5 What do infectious disease clinicians need to ...
6 The public health burden attributable to asthm...
7 The progress of the U.S. smallpox vaccination ...
8 Influenza is linked with increased risk for ad...
9 Read about the new drug approval for Hepsera f...

```

```

                                clean_text
0 infectious disease october bartlett review lat...
1 influenza vaccination pregnancy enough family ...
2 bacterial meningitis deadly preventable diseas...
3 hepatitis vaccination patient chronic liver di...
4 emerge infection john bartlett report late dat...
5 organism drug test guideline really need know ...
6 asthma influenza vaccination public health bur...
7 answer smallpox vaccination question progress ...
8 flu vaccine cardiovascular prevention elderly ...
9 october read drug approval hepsera chronic hep...

```

```

before = len(df)
df = df[df["clean_text"].str.strip().ne("")]<div data-bbox="111 682 400 715" data-label="Text">


```

Removed 0 empty documents.
Documents remaining: 366

```


```

```

all_tokens = [token for tokens in df["tokens"] for token in tokens]
word_counts = Counter(all_tokens)

```

```

freq_df = pd.DataFrame(
    word_counts.most_common(30),
    columns=["word", "frequency"]
)

```

```

display(freq_df.head(20))

```

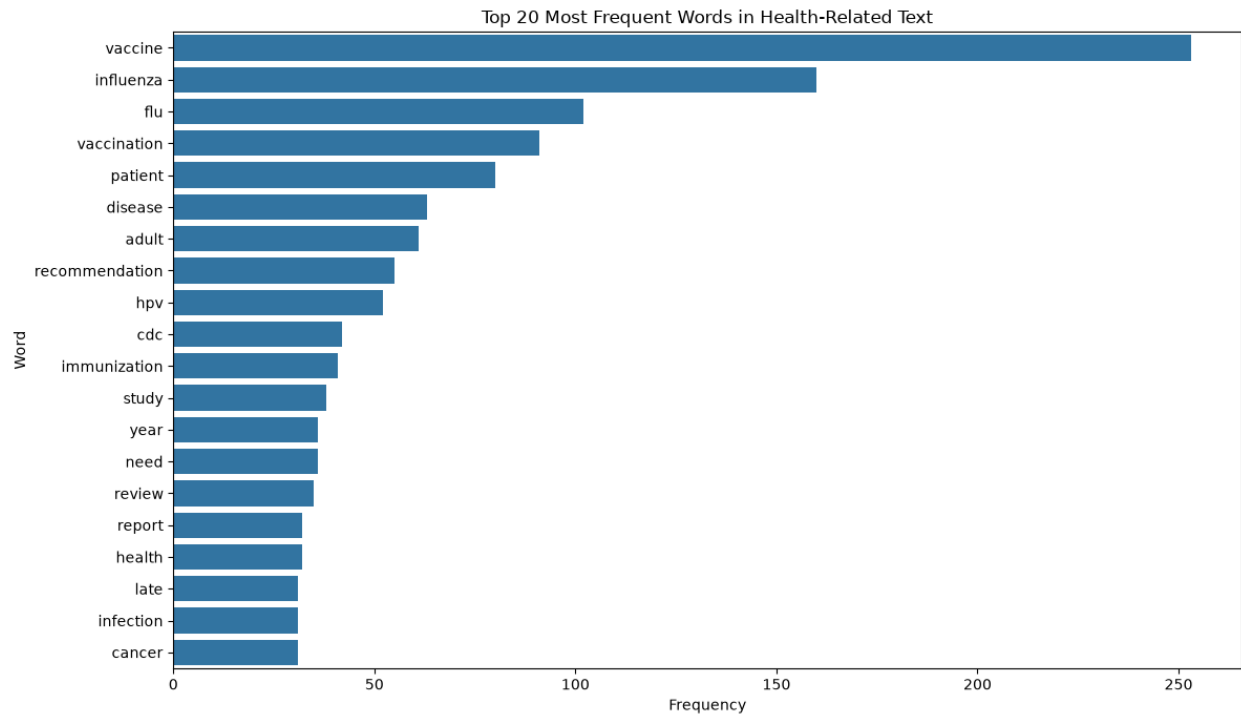
	word	frequency
0	vaccine	253
1	influenza	160
2	flu	102
3	vaccination	91
4	patient	80
5	disease	63
6	adult	61
7	recommendation	55
8	hpv	52
9	cdc	42
10	immunization	41
11	study	38
12	year	36
13	need	36
14	review	35
15	report	32
16	health	32
17	late	31
18	infection	31
19	cancer	31

```

top20 = word_counts.most_common(20)
words = [x[0] for x in top20]
counts = [x[1] for x in top20]

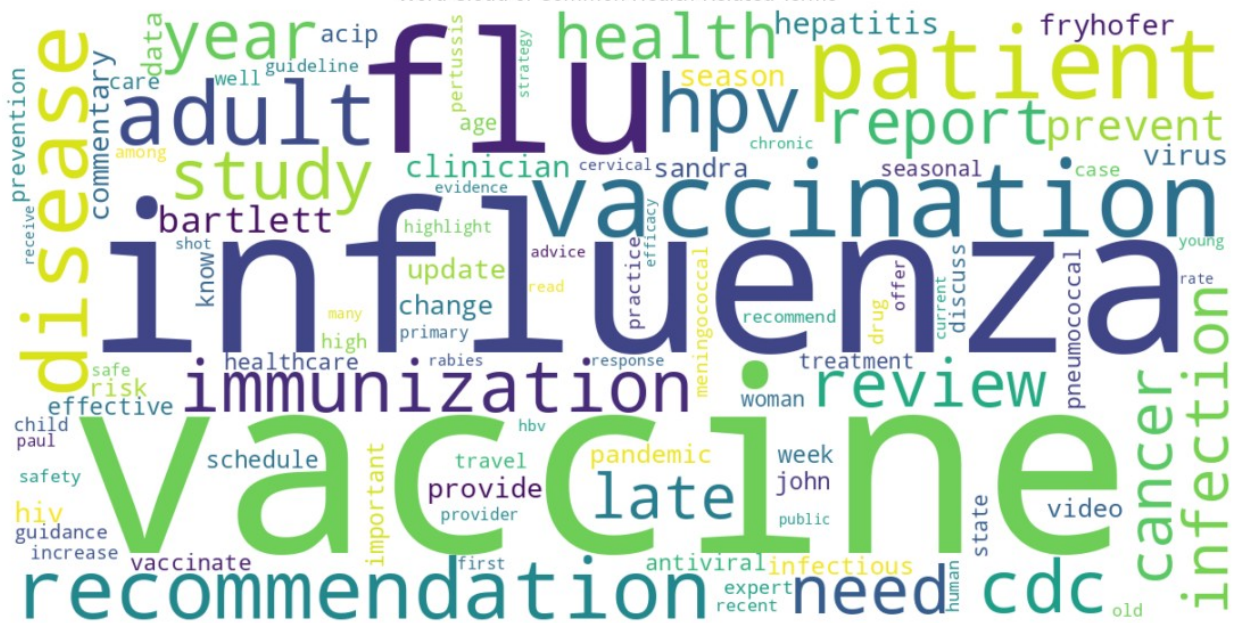
plt.figure(figsize=(12, 7))
sns.barplot(x=counts, y=words)
plt.title("Top 20 Most Frequent Words in Health-Related Text")
plt.xlabel("Frequency")
plt.ylabel("Word")
plt.tight_layout()
plt.show()

```



```
wordcloud = WordCloud(  
    width=1200,  
    height=600,  
    background_color="white",  
    max_words=100,  
    collocations=False  
)  
.generate(" ".join(all_tokens))  
  
plt.figure(figsize=(15, 7))  
plt.imshow(wordcloud, interpolation="bilinear")  
plt.axis("off")  
plt.title("Word Cloud of Common Health-Related Terms")  
plt.show()
```

Word Cloud of Common Health-Related Terms



```
NUM_TOPICS = 5
NUM_TOP_WORDS = 5

vectorizer = CountVectorizer(
    max_df=0.95,
    min_df=2,
    stop_words="english"
)

doc_term_matrix = vectorizer.fit_transform(df["clean_text"])

print("Document-term matrix shape:", doc_term_matrix.shape)
print("Vocabulary size:", len(vectorizer.get_feature_names_out()))

Document-term matrix shape: (366, 635)
Vocabulary size: 635

lda_model = LatentDirichletAllocation(
    n_components=NUM_TOPICS,
    random_state=42,
    learning_method="batch",
    max_iter=20
)

lda_model.fit(doc_term_matrix)

feature_names = vectorizer.get_feature_names_out()

def display_topics(model, feature_names, n_top_words=5):
    topic_rows = []
```

```

for topic_idx, topic in enumerate(model.components_):
    top_indices = topic.argsort()[:-n_top_words - 1:-1]
    top_words = [feature_names[i] for i in top_indices]
    topic_rows.append({
        "Topic": f"Topic {topic_idx + 1}",
        "Top Words": ", ".join(top_words)
    })
return pd.DataFrame(topic_rows)

topic_table = display_topics(
    lda_model,
    feature_names,
    n_top_words=NUM_TOP_WORDS
)

display(topic_table)

```

	Topic	Top Words
0	Topic 1	patient, immunization, vaccination, hepatitis,...
1	Topic 2	vaccine, hpv, adult, vaccination, recommendation
2	Topic 3	flu, vaccine, disease, patient, shot
3	Topic 4	vaccine, influenza, disease, flu, review
4	Topic 5	influenza, vaccine, flu, vaccination, season

```

topic_distribution = lda_model.transform(doc_term_matrix)
df["dominant_topic"] = topic_distribution.argmax(axis=1) + 1

print("Documents by dominant topic:")
display(df["dominant_topic"].value_counts().sort_index().rename("document_count").to_frame())

display(
    df[["headlines", "clean_text", "dominant_topic"]].head(15)
)

```

Documents by dominant topic:

dominant_topic	document_count
1	75
2	83
3	51
4	69
5	88

	headlines \
0	Infectious Diseases: October 30, 2004
1	Influenza Vaccination in Pregnancy
2	Bacterial Meningitis: A Deadly But Preventable...
3	Hepatitis Vaccination in Patients With Chronic...
4	Emerging Infections

```

5 New Organisms, New Drugs, New Tests, New Guide...
6     Asthma and Influenza Vaccination
7     Some Answers to Smallpox Vaccination Questions
8 Flu Vaccine for Cardiovascular Prevention in t...
9     October 2002
10 Special Report: Adult Vaccines -- New ACIP Rec...
11     Condoms Prevent HSV-2 Acquisition
12 Influenza Vaccination Levels - United States, ...
13 Response Plans Prepare for Avian Influenza Out...
14 siRNA in Prophylactic and Therapeutic Regimens...

```

	clean_text	dominant_topic
0	infectious disease october bartlett review lat...	4
1	influenza vaccination pregnancy enough family ...	2
2	bacterial meningitis deadly preventable diseas...	3
3	hepatitis vaccination patient chronic liver di...	1
4	emerge infection john bartlett report late dat...	4
5	organism drug test guideline really need know ...	1
6	asthma influenza vaccination public health bur...	1
7	answer smallpox vaccination question progress ...	1
8	flu vaccine cardiovascular prevention elderly ...	3
9	october read drug approval hepsera chronic hep...	3
10	special report adult vaccine acip recommendati...	2
11	condom prevent hsv acquisition condom prevent ...	4
12	influenza vaccination level unite state improv...	4
13	response plan prepare avian influenza outbreak...	4
14	siRNA prophylactic therapeutic regimen sars de...	1

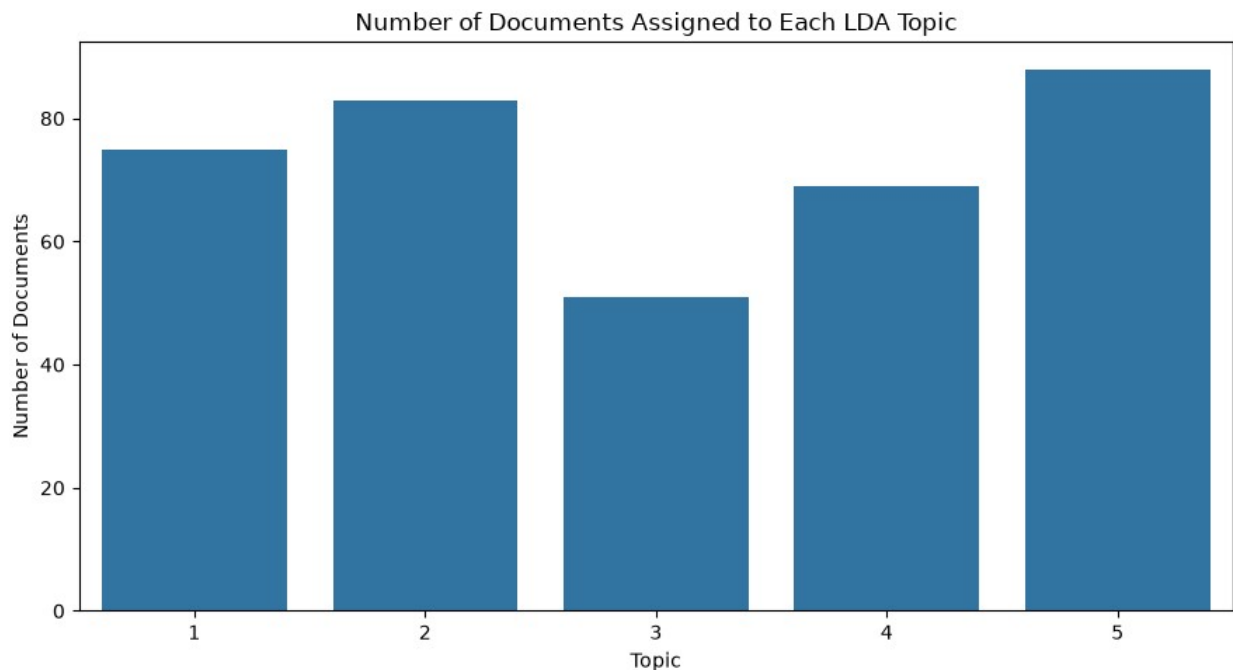
```

topic_counts = df["dominant_topic"].value_counts().sort_index()

plt.figure(figsize=(9, 5))
sns.barplot(x=topic_counts.index.astype(str), y=topic_counts.values)
plt.title("Number of Documents Assigned to Each LDA Topic")
plt.xlabel("Topic")
plt.ylabel("Number of Documents")

```

```
plt.tight_layout()
plt.show()
```



```
import pyLDAvis
import pyLDAvis.lda_model
```

```
pyLDAvis.enable_notebook()
```

```
lda_vis = pyLDAvis.lda_model.prepare(
    lda_model,
    doc_term_matrix,
    vectorizer,
    mds="tsne"
)
```

```
display(lda_vis)
```

PreparedData(topic_coordinates=				x	y	topics
cluster	Freq					
topic						
4	-82.324089	26.736567	1	1	23.762863	
1	26.709929	53.343029	2	1	22.801242	
0	-35.260979	76.943665	3	1	20.225345	
3	-30.501713	-14.586076	4	1	19.802713	
2	36.094444	-14.746188	5	1	13.407836	topic_info=
Term	Freq	Total	Category	logprob	loglift	
293	influenza	142.000000	142.000000	Default	30.0000	30.0000
262	hpv	47.000000	47.000000	Default	29.0000	29.0000

609	vaccine	224.000000	224.000000	Default	28.0000	28.0000
219	flu	90.000000	90.000000	Default	27.0000	27.0000
13	adult	55.000000	55.000000	Default	26.0000	26.0000
...	...	...	...	...	...	...
293	influenza	9.803630	142.165342	Topic5	-4.1717	-0.6649
278	important	4.794413	18.443555	Topic5	-4.8870	0.6621
630	year	5.196249	32.636436	Topic5	-4.8065	0.1718
493	risk	4.503157	19.359494	Topic5	-4.9497	0.5509
244	healthcare	4.145757	17.589428	Topic5	-5.0324	0.5641

[305 rows x 6 columns], token_table= Topic Freq Term

term			
1	4	0.906944	abroad
3	2	0.752050	achieve
4	2	0.410724	acip
4	3	0.564745	acip
7	4	0.809637	activity
...	...	...	...
630	3	0.122562	year
630	4	0.122562	year
630	5	0.153203	year
632	2	0.951908	young
634	5	0.902202	zoster

[446 rows x 3 columns], R=30, lambda_step=0.01, plot_opts={'xlab': 'PC1', 'ylab': 'PC2'}, topic_order=[5, 2, 1, 4, 3])

OUTPUT_PATH = "health_headlines_cleaned_with_topics.csv"

```
save_df = df.copy()
save_df["tokens"] = save_df["tokens"].apply(lambda x: " ".join(x))
save_df.to_csv(OUTPUT_PATH, index=False)
```

print(f"Saved cleaned dataset to: {OUTPUT_PATH}")

Saved cleaned dataset to: health_headlines_cleaned_with_topics.csv